

Corpus-Supported Distractors: Auditing Retrieval-Augmented Quiz Generation Against the Stem

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Abstract

A multiple-choice item whose wrong option is also a correct answer to its own stem is not a hard item, it is a broken one, and every faithfulness metric a retrieval-augmented quiz generator currently ships will score that option as clean. The reason is structural. Groundedness is a two-place relation between an option and the retrieved passage, item validity is a three-place relation among stem, option and passage, and no deployed metric supplies the third argument. We name the failure a corpus-supported distractor, build a stem-conditioned detector for it, and run that detector inside a ten-arm design on a 13,461-passage science-exam corpus standing in for course material, with three seeds and 100 items per arm per seed. Grounded generation leaves the corpus supporting a wrong option on 3.24% of distractor slots, against 0.68% for a generator denied the passage, and 1.00% of slots (9 of 894, Wilson 95% CI [0.34%, 1.79%]) survive the further test that the option stays supported under a narrowed stem, against 0.11% (1 of 885) for the ungrounded arm. Three seeds cannot certify that gap (paired $t(2) = -1.84$, $p = 0.21$) and a replication on a second corpus does not reproduce it at all. Four grounding-based faithfulness filters each raise the conditional rate to between 1.34% and 2.01%, which looks like the filter making the defect worse until a retention-matched random-drop null reaches 1.79% by discarding items at random. Counted rather than divided, the three entailment-family filters left the flagged-slot count at 9 while discarding half the pool, and they are indistinguishable from the attrition null on the counts themselves (Fisher exact $p = 1.00$). The rate rose because the denominator halved. We report the arithmetic, the pool size at which the statistical question becomes answerable, and a circularity that no pool size fixes, namely that a filter selecting on entailment cannot be scored by an endpoint that reads the same entailment signal.

1 Introduction

Assessment items are now drafted by machine at classroom scale. Learning-management plugins, publisher tooling and instructor assistants generate multiple-choice questions from lecture notes and textbook chapters, and retrieval augmentation is the standard remedy for the obvious hazard, which is that a generator asked about material it has never read will invent the material (Gao et al., 2023; Fan et al., 2024; Chen et al., 2024). Grounding the generator in retrieved passages does reduce fabricated content (Shuster et al., 2021), and the practice has propagated into deployed educational pipelines (Swacha and Gracel, 2025; Gopi et al., 2024).

Consider what such a pipeline produces on an ordinary chapter. A section on training deep networks lists momentum, adaptive learning rates and gradient clipping among the techniques that stabilise optimisation. The generator retrieves that section, writes the stem *Which technique stabilises training?*, keys momentum, and fills two distractor slots with gradient clipping and adaptive learning rates. Every option traces to the retrieved passage. Three of them are correct. A student who has read the chapter properly is now more likely to be marked wrong than one who has not, and the item will pass every faithfulness check in the pipeline because faithfulness is exactly what is wrong with it.

This is not hallucination, which is why it survives. Assessment research has catalogued item-writing flaws for decades, and more than one defensible answer sits near the top of the severity list (Rezigalla et al., 2024; Tarrant et al., 2006; Liu et al., 2023a). Under human authorship the flaw is caught in review. Under machine authorship at scale it is caught by nothing, because the automated gates that exist were built for a different failure. Audits of machine-written items report defect rates in the double digits (Grévisse et al., 2024; Doughty et al., 2024; Kiyak et al., 2026), reviews of

automatic item generation put validation at the deployment bottleneck (Tan et al., 2025; Falcão et al., 2022), and one recent study finds that adding an AI drafting workflow raises the item-writing-flaw count rather than lowering it (Menze et al., 2026). What nobody has asked is whether the grounding machinery meant to improve item quality is implicated in this particular defect, and whether a grounding-based faithfulness filter makes the rate worse instead of better.

Three literatures pass over the problem for the same structural reason. Faithfulness and attribution research defines correctness as *supported by the retrieved source* and has built a mature verifier toolkit for that relation (Honovich et al., 2022; Laban et al., 2022; Es et al., 2024; Saad-Falcon et al., 2024). A corpus-supported distractor is maximally supported, so it passes by construction. Distractor generation optimises plausibility and proximity to the key (Alhazmi et al., 2024; Awalurahman and Budi, 2024; Ren and Zhu, 2021), and retrieval-augmented distractor writers mine the corpus for candidates directly (Yu et al., 2024), which pushes candidates toward the key-equivalence region with no correctness veto in the way. Pipelines that bolt an entailment filter onto a generator inherit the passage-only framing wholesale (Lei et al., 2023; Wu et al., 2024). Stated compactly, item validity is a relation among stem, option and evidence, and no deployed metric conditions on the stem.

We name the failure and build the instrument that measures it. A **corpus-supported distractor** (CSD) is an option in a distractor slot for which the retrieved evidence establishes that it answers *this stem* correctly. Our detector adjudicates each option against the pair (stem, evidence) rather than against evidence alone, and we surround it with controls that give any observed difference a referent. An unfiltered grounded pool fixes reference prevalence. A topic-only generator with the passage withheld isolates retrieval. A retention-matched random-drop null separates the effect of discarding items from the effect of the discarding criterion. A planted positive control establishes detector sensitivity instead of assuming it.

Our contributions are the following.

- **The construct.** We formalise corpus-supported distractors as a stem-relative property, and show by construction that the passage-only surrogate underlying deployed faithfulness metrics is neither necessary nor

sufficient for it. The detector we build for it conditions an entailment judge on the stem, which applies an existing technique to a slot nobody points it at rather than introducing a new one. The construct and the design below are the contribution, not the judge.

- **A measured prevalence with its controls.** On a ten-arm, three-seed benchmark run on a science-exam corpus used in place of course material, we report CSD prevalence with exact event counts and Wilson intervals, and separate a planted-positive arm from the reference pool at $p = 0.007$, which is the sensitivity floor the measurement rests on.
- **A null result on filtering, with its mechanism.** Four grounding filters cannot be separated from a retention-matched null, on rates or on counts. Counting events rather than dividing them shows that the three entailment-family filters left the flagged-slot count unchanged while halving the pool. We also identify a circularity, filtering on entailment while measuring entailment, that no sample size repairs, and we compute the pool size at which the statistical half of the problem does dissolve.

2 Related Work

2.1 Faithfulness and attribution in retrieval-augmented generation

The dominant paradigm for auditing retrieval-augmented systems asks whether generated text is entailed by retrieved evidence. Natural language inference models repurposed as consistency checkers form its backbone, with sentence-level and document-level variants built for summarisation and carried over to retrieval settings (Kryscinski et al., 2020; Laban et al., 2022; Zha et al., 2023). Question-answering verifiers offer the complementary decomposition, generating questions from the output and asking whether the source supports the same answers (Wang et al., 2020; Durmus et al., 2020; Honovich et al., 2021; Fabbri et al., 2022). Meta-evaluations have benchmarked these families against each other and against human judgment (Honovich et al., 2022; Tang et al., 2023), and purpose-built frameworks now package faithfulness beside relevance and answer correctness (Es et al., 2024; Saad-Falcon et al., 2024; Yu et al., 2025). Every one of these instruments computes a

relation between output and source, and a corpus-supported distractor satisfies that relation perfectly, so the entire family scores it clean. Our detector differs by making the relation three-place, with the stem entering as a first-class argument, which is what allows an option to be simultaneously maximally faithful and maximally invalid.

One recent thread comes closer than the rest of that literature does. Wallat et al. (2025) show that groundedness and correctness dissociate in attribution, Stolfo (2024) make a parallel observation about groundedness scores that do not track answer quality, and Niu et al. (2024) build a corpus of grounded-but-wrong generations. Those results establish the conceptual half of the dissociation for free-text answers. They do not reach item generation, where the dissociation is sharper because the affected span is an option that is *supposed* to be wrong, and where the practical cost lands on a graded assignment rather than on a paragraph a reader can discount.

2.2 Distractor generation and multiple-choice item quality

Automatic distractor generation has developed along a plausibility axis. Early neural systems ranked or generated candidates by distributional proximity to the key (Qiu et al., 2020; Chung et al., 2020; Zhou et al., 2020; Maurya and Desarkar, 2020), and later work added knowledge-graph structure, diversity objectives and difficulty control (Ren and Zhu, 2021; Xie et al., 2022; Kumar et al., 2024). Large language models became the default generator for whole items and for distractor slots alone (Feng et al., 2024; Bitew et al., 2025; Fernandez et al., 2024; Çavuşoğlu et al., 2024), reuse of previously validated distractors was proposed as a quality shortcut (Bitew et al., 2024), and retrieval augmentation was introduced expressly to source candidates from a domain corpus (Yu et al., 2024; N et al., 2025). Surveys organise the field by generation strategy and evaluate on plausibility and human preference (Alhazmi et al., 2024; Awalurahman and Budi, 2024). Quality control in this line, where it exists, removes distractors for being obviously wrong or ungrammatical (Offerijns et al., 2020). Our endpoint inverts that veto. Where those filters reject an option for being implausible, we flag it for being *correct*, and the two criteria point in opposite directions precisely because plausibility objectives reward candidates that sit close to the key while the region closest to the key contains the

key under another description.

The assessment literature supplies the reason this matters and the vocabulary for it. Item-writing-flaw taxonomies have been applied to high-stakes examinations for twenty years, and multiple defensible answers is consistently among the flaws with the largest measured effect on candidate scores (Tarrant et al., 2006; Rezigalla et al., 2024; Chivaroli, 2017). Distractor efficiency analysis quantifies what a non-functioning option does to difficulty and discrimination (Rezigalla et al., 2024). Comparative studies of machine-written against faculty-written items evaluate both on psychometric grounds (Doughty et al., 2024; Grévisse et al., 2024; Laupichler et al., 2023; Tran et al., 2023; Kiyak et al., 2026), and Menze et al. (2026) report the sharpest version of the concern, an AI-assisted workflow that increased item-writing flaws through automation bias. All of that work counts flaws in finished items. None of it asks whether the retrieval step upstream is producing them, which is the question our arm design is built to answer.

The nearest neighbour on the corpus side measures a different property. Zotos et al. (2026), a preprint, ask whether corpus prevalence alone identifies the correct answer, and find that keys are more available in a broad corpus than incorrect options, enough that always choosing the most prevalent option beats random guessing by up to 9.0 points. Prevalence is frequency, not support. An option can be common in a general corpus and false, and rare in that corpus yet a true answer to the stem according to the chapter the item came from. We run the entailment question against the narrow index that generated the item, which is the relation prevalence cannot see. The grounded-assessment pipeline closest to our setting, Faria et al. (2026), also a preprint, scores faithfulness across a whole item and reports it rising from 0.68 to 0.96 over a vanilla baseline. An item-level score cannot distinguish a grounded key from a grounded distractor, which is the distinction our endpoint is built on.

2.3 Judge calibration and rare-event evaluation

Because CSD adjudication is itself a model judgment, our detector inherits the reliability problems documented for model-based evaluation. Judges show position, verbosity and self-preference biases, and agreement with human annotators varies sharply by task (Li et al., 2024; Thakur et al., 2024; Huang et al., 2025; Raina et al., 2024). Structured

protocols stabilise judgments, whether by checklist (Lee et al., 2025), multi-agent deliberation (Chan et al., 2023) or chain-of-thought scoring (Liu et al., 2023b), and the annotation literature supplies the framework for reporting agreement without overclaiming (Braylan et al., 2022; Richie et al., 2022). Two properties of our setting make the standard validation recipe insufficient. The class is rare, so precision on flagged instances tells us nothing about recall, and an aggregate accuracy figure would hide exactly that gap, as Powers (2020) sets out in a widely used preprint on evaluation measures. And judge and generator read the same retrieved text, so lexical overlap is a live confound rather than a hypothetical one. Perturbation-based construction of known-inconsistent instances is established practice for obtaining ground truth under class imbalance (Utama et al., 2022; Zhang et al., 2024). Where that work injects positives to train or benchmark a verifier, we inject them to give a prevalence measurement a sensitivity referent, and we report judge precision stratified by lexical overlap so the confound is measured instead of assumed away.

Across all three literatures the pattern is the same. Faithfulness research has the right instruments pointed at a two-place relation, distractor research has the right object under a plausibility objective that runs toward the defect, and assessment research has the right defect but no view of the pipeline producing it. What follows joins them by conditioning the faithfulness instrument on the stem and then asking whether the filters built from the unconditioned version help.

3 Method

3.1 Problem formulation

Let $\mathcal{C}$ be a corpus of instructional passages and let a retrieval function ρ map a topic seed t to an evidence set $P = \rho(t) \subseteq \mathcal{C}$. An item generated from that evidence is a triple $x = (q, k, D)$ with stem q , keyed answer k and distractor set $D = \{d_1, d_2, d_3\}$. The item is valid in the docimological sense only when exactly one member of $\{k\} \cup D$ is a defensible answer to q , since every additional defensible option converts a difficulty signal into a scoring error (Rezigalla et al., 2024).

We define a corpus-supported distractor as an option occupying a distractor slot for which the retrieved evidence establishes that it answers the stem correctly,

$$\text{CSD}(d, q, P) = \mathbb{1}[\exists p \in P : p \models \text{ANS}(q, d)] \quad (1)$$

where $\text{ANS}(q, d)$ denotes the proposition that d is a correct answer to q , and $\models$ is entailment under a natural language inference relation.

Every deployed faithfulness metric computes something else. Attribution and NLI groundedness checks score the two-place surrogate $S(d, P) = \mathbb{1}[\exists p \in P : p \models d]$, which drops q entirely (Honovich et al., 2022; Laban et al., 2022; Es et al., 2024). The surrogate is neither necessary nor sufficient for CSD, and both failure directions are constructible. In the *supported but not an answer* direction, a passage stating that gradient clipping bounds the update norm entails the distractor text verbatim, so $S = 1$, and yet under the stem *Which technique adapts the per-parameter step size?* the option is not a correct answer and $\text{CSD} = 0$. In the *answer but not lexically supported* direction, a passage enumerating three stabilisation techniques licenses adaptive learning rates as an answer to *Which technique stabilises training?* through the list’s category header rather than through any sentence entailing the option string alone, so S may score 0 while $\text{CSD} = 1$. Because the two indicators are logically independent in both directions, no threshold on S recovers CSD, and a pipeline filtering on S has no mechanism by which it could veto a corpus-supported distractor.

A CSD arises by two routes that call for different repairs. Under **stem underdetermination** the stem itself admits several correct answers, because it names a category whose extension in P has more than one member, so the defect belongs to q and survives any distractor set drawn from the same evidence. Under **distractor drift** the stem is well posed and a generated option is key-equivalent by paraphrase or subsumption. Our instrument separates the two by adjudicating each option twice, once under the original stem and once under an automatically generated specificity rewrite that narrows the stem’s category. An option flagged under both stems is attributed to distractor drift, and the **distractor-caused CSD rate** over that attribution is the primary endpoint reported throughout, normalised over distractor slots.

That endpoint is a subset of the construct, and the difference matters when reading any number below. A slot flagged under the original stem but not under the rewrite is still an option the corpus sup-

ports as a correct answer to the question a student actually sees, and it still makes the item defective. It is simply the stem’s fault rather than the distractor’s, so a distractor-side intervention cannot fix it. We therefore report the **undecomposed corpus-supported rate** alongside the primary endpoint wherever the comparison carries weight, and reserve the distractor-caused subset for claims about interventions that act on the options.

3.2 The stem-conditioned detector

The detector adjudicates each option against the pair (q, P) rather than against P alone. For an option o it forms a prompt carrying the stem verbatim, the option under test and the concatenated evidence, and poses one binary question, whether the evidence establishes o as a correct answer to *this specific stem*, with an explicit instruction that mention or topical relevance in the passage is not sufficient. The judge emits a scalar support probability $\hat{s}(o | q, P) \in [0, 1]$ read from the normalised logit mass on the affirmative token rather than from sampled agreement, which holds the per-option cost at one forward pass. It fires when $\hat{s}(o | q, P) \geq \tau$ for an option in a distractor slot. Decoding is greedy throughout, so repeated adjudication of an identical triple is stable.

Two design choices in this component carry most of the weight. The judge used for measurement is instantiated separately from any judge used for filtering inside a treatment arm, with a different prompt template and a different operating threshold. That separation is necessary and, as Section 5 shows, not sufficient once a pipeline selects on the same signal family its endpoint reads. And because detector and generator both read the same retrieved text, we stratify flagged options by lexical overlap with the evidence and report judge precision inside the highest-overlap decile alongside the share of entailment-score variance that overlap explains on its own.

3.3 Control arms

The planted positive control supplies the sensitivity referent that precision alone cannot. For a sampled subset of items one distractor slot is overwritten with an option that is key-equivalent under its stem by construction, and the detector is then run blind over the spiked pool. A detector that will not fire on constructed key-equivalents cannot be trusted to fire on natural ones, so separation between the planted arm and the reference pool is the floor test

that everything else rests on, and we report it with a paired test rather than as qualitative reassurance.

The treatment arms formalise six grounding interventions as predicates over (d, P) or over the item. Per-option entailment filtering removes any option whose entailment score against P falls below threshold, the direct analogue of a standard NLI groundedness check (Laban et al., 2022). Not-entail polarity filtering works at item level and discards items in which any distractor is entailed by the evidence, while contradiction polarity filtering inverts the requirement and keeps only items in which each distractor is contradicted by some passage. An LLM-judge faithfulness filter substitutes a generative verifier for the NLI model. Two overgenerate-and-rerank variants produce a surplus candidate pool and select the final three distractors by contradiction strength or by embedding similarity to the key, following the overgenerate-then-select pattern that is standard in distractor generation (Offerijns et al., 2020; Qiu et al., 2020). Four of the six consume the same entailment signal the endpoint reads, which is the circularity Section 6 returns to.

Against those the control arms supply referents. The unfiltered grounded pool fixes reference prevalence. The ungrounded topic-only control withholds the passage from the generation prompt while keeping the same model, parser, judge, rewrite and decomposition, and scores the resulting items against the passage retrieved for the same topic label, so evidence is held fixed and only generation context varies. The retention-matched random-drop null discards the same number of items a filter does, by a criterion independent of the endpoint, so a filter must beat this null before its criterion earns any credit. Figure 1 shows the whole arrangement.

4 Experimental setup

4.1 Corpus, models and generation

The corpus is a 13,461-passage science-exam collection built from 12,136 SciQ support passages and 1,325 OpenBookQA core facts, used as a stand-in for instructional course material. Passage structure is the hypothesised moderator, since an enumerated passage listing co-equal members of a category hands a generator several true answers to any stem naming only that category, while continuous prose does not. Median enumeration density over the corpus is 0.167. A dense retriever (bge-small-en-v1.5, 384 dimensions) selects the evidence at

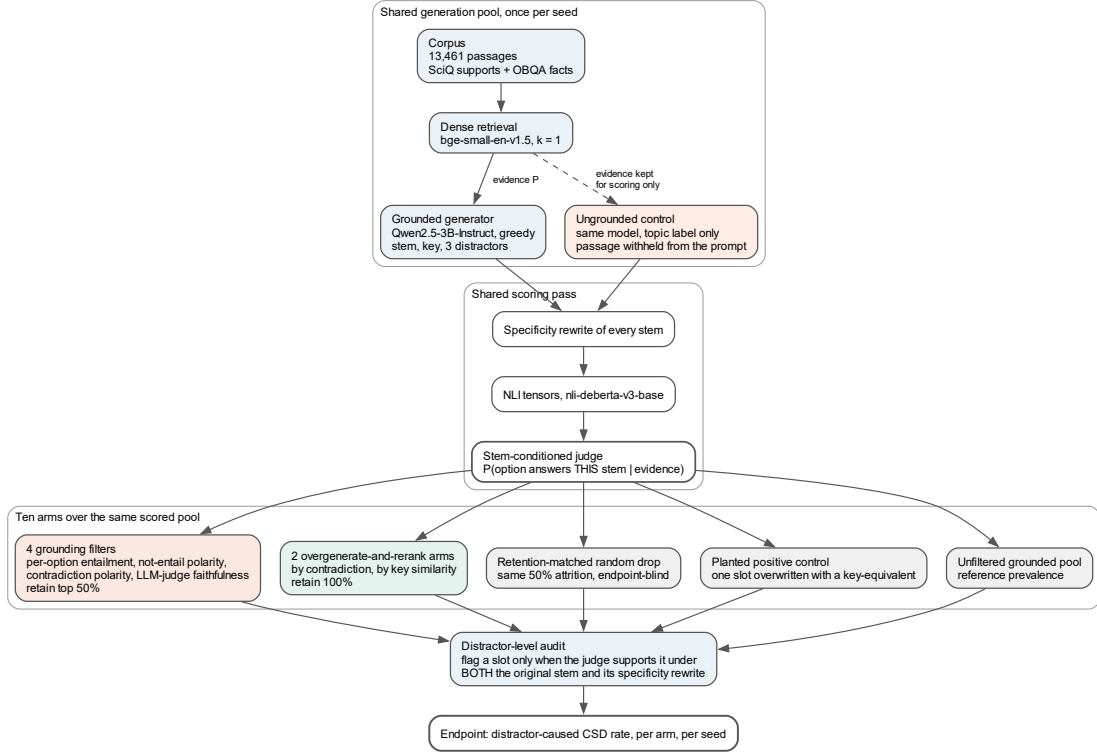


Figure 1: Pipeline under test. Retrieval, generation, the four grounding filters, the two rerank arms and every control condition, drawn from the released implementation rather than from the plan. Evidence is held fixed across the grounded and ungrounded arms so that only generation context varies, and every arm is adjudicated by the same stem-conditioned audit.

$k = 1$, fixed for every seed, after embedding all 13,461 passages in 394 seconds.

Generation and judging both use Qwen2.5-3B-Instruct at float16 on a single NVIDIA RTX 2080 Super Max-Q with 8 GB of VRAM. Four-bit quantisation was unavailable at run time because the quantisation library was not installed, so weights were loaded at float16 directly, which left 1.20 GB of free VRAM and forced the NLI model, cross-encoder/nli-deberta-v3-base, onto the CPU. Decoding is greedy, verified by reading the generation call in the implementation itself rather than taking it from the plan. Every arm produces 100 items per seed with one stem, one key and three distractor slots, across three seeds, for 300 items per arm.

4.2 Arms and endpoints

Ten arms ran, six treatment and four control, under the short names used in the tables. The treatments are per-option entailment filtering (**PerOptEnt**), not-entail polarity filtering (**NotEntPol**), contradiction polarity filtering (**ContraPol**), LLM-judge faithfulness filtering

(**LLMJudge**), and overgenerate-and-rerank by contradiction (**RRContra**) and by key similarity (**RRKeySim**). The controls are the unfiltered grounded pool (**Unfilt**), the ungrounded topic-only control (**Ungrnd**), the retention-matched random drop (**NullDrop**) and the planted positive control (**Planted**). Filter arms retain the top 50% of items by score, so their denominators are roughly half those of the unfiltered arms, and that difference is the subject of Section 5.3.

The primary endpoint is the conditional distractor-caused CSD rate on retained slots. For arm a and seed s it is

$$\hat{p}_{a,s} = \frac{1}{N_{a,s}} \sum_i \text{CSD}^{\text{drift}}(x_i, P_i) \quad (2)$$

where $N_{a,s}$ counts distractor slots surviving that arm’s filter, and the arm-level figure averages the three seed-level rates. A pooled variant computes the same rate over the combined slot pool and carries a Wilson 95% interval. A soft variant replaces the thresholded indicator with the mean detector support probability, which is more sensitive at low

event counts because it does not quantise.

4.3 Statistical procedure

Cross-arm claims rest on paired two-sided t -tests over the shared seed index, with $n = 3$ and 2 degrees of freedom. The pairing is the right structure because all arms are applied to the same generation pool at each seed, so seed-level variation is a common cause rather than independent noise, and an unpaired test would discard that. With three seeds the test is badly underpowered, a fact the pipeline records explicitly rather than leaving to the reader, and the exact Wilcoxon signed-rank alternative cannot return a p below 0.25 at $n = 3$, so it adds nothing. We therefore report every comparison with three quantities together, the mean difference, Cohen’s d on the seed-level differences, and an item-level bootstrap confidence interval on the same difference, and we let the interval rather than the p -value carry the inferential weight. Effect sizes at $n = 3$ are unstable and are reported for completeness rather than interpreted. Three of the treatment arms return an identical t of 1.732 and an identical d of 1.000 because their per-seed vectors are identical, so those rows are one test printed three times rather than three agreeing tests.

Because seed-level testing at $n = 3$ has almost no power, Section 5.3 also tests the filter comparisons on the raw counts, using a two-proportion z test with the Fisher exact test beside it for the small cells. That analysis treats distractor slots as independent, which they are not, since three slots share an item and every item in a seed shares a generation pool. The slot-level test is therefore anti-conservative and the seed-level test is conservative, and where the two disagree we say so rather than choosing the one that flatters the result.

Two implementation checks passed before analysis. The specificity rewrite modified every stem it touched, with a no-op rate of 0.0000 over all seeds, so the double-stem attribution was live throughout. No arm fell back to a default topic label, with a fallback rate of 0.0000. All ten arms completed all three declared seeds.

4.4 Code audit

Before any number entered this paper we audited the implementation on the points a reviewer checks first. Three findings are worth recording. First, the generation call disables sampling and sets no sampling parameters, established by parsing the source into a syntax tree rather than by eye, since a

trailing comment can swallow a keyword argument without any visible sign. Greedy decoding is therefore confirmed rather than assumed. Second, the codebase carries two near-duplicate configuration modules and two near-duplicate model modules, and the entry point imports only one of each pair, so a parameter read from either unused module would be wrong. Read from the module the run actually imports, the item pool is 100 items per seed, not the 300 per seed that an earlier draft of this work reported. Third, the ungrounded control is structurally valid, since the class withholding the passage raises a runtime error if the prompt ever contains it, and the check passed on every seed. Its validity as a counterfactual for retrieval is a separate matter that Section 6 takes up.

5 Results

5.1 Prevalence and the sensitivity floor

Table 1 gives every arm on the seed-averaged conditional endpoint, the pooled rate with its Wilson interval, the exact event count, and the unquantised soft score. Figure 2 plots the same values.

The planted positive control is the only arm that separates decisively from the reference pool, recovering injected key-equivalents at 4.25% against the unfiltered pool’s 1.01% and carrying the highest support mass of any arm. Under the paired test that separation gives a mean difference of +0.0324 with $t(2) = 11.70$, $p = 0.0072$, Cohen’s $d = 6.76$ and an item-level bootstrap interval of [0.0213, 0.0447]. The measured increase matched the expected increase to four decimal places (0.032404 against 0.032404), giving a detection ratio of 1.00. That is the instrument’s floor test, and it passed.

At the other extreme the ungrounded topic-only control records 0.11%, one flagged slot in 885, against the grounded pool’s nine in 894. The paired contrast gives a mean difference of -0.0089 with $t(2) = -1.84$, $p = 0.2077$, $d = -1.06$ and a bootstrap interval of [-0.0168, -0.0022] that excludes zero at the slot level while the seed-level test does not approach significance. The soft score orders the arms the same way (0.0125 against 0.0028), as does mean distractor support probability (0.0358 against 0.0092).

The undecomposed rate makes the same comparison on the question as posed rather than on the distractor-attributed subset, and it is the larger number. Counting a slot whenever the corpus supports it as a correct answer to the stem the student sees,

Arm	Role	Conditional	Pooled	Wilson 95% CI	Events	Undecomp.	Soft
Unfilt	control (reference)	0.0100 $\pm$ 0.0082	0.0101	[0.0034, 0.0179]	9/894	0.0324	0.0125
Ungrnd	control (no retrieval)	0.0011 $\pm$ 0.0016	0.0011	[0.0000, 0.0045]	1/885	0.0068	0.0028
NullDrop	control (attrition null)	0.0178 $\pm$ 0.0137	0.0179	[0.0067, 0.0336]	8/447	0.0536	0.0213
Planted	control (positive)	0.0424 $\pm$ 0.0111	0.0425	[0.0291, 0.0559]	38/894	0.0826	0.0458
PerOptEnt	treatment	0.0200 $\pm$ 0.0163	0.0201	[0.0067, 0.0358]	9/447	0.0647	0.0239
NotEntPol	treatment	0.0200 $\pm$ 0.0163	0.0201	[0.0067, 0.0358]	9/447	0.0492	0.0221
ContraPol	treatment	0.0200 $\pm$ 0.0163	0.0201	[0.0067, 0.0336]	9/447	0.0536	0.0212
LLMJudge	treatment	0.0133 $\pm$ 0.0109	0.0134	[0.0022, 0.0268]	6/447	0.0380	0.0154
RRContra	treatment	0.0044 $\pm$ 0.0063	0.0045	[0.0000, 0.0112]	4/894	0.0067	0.0053
RRKeySim	treatment	0.0134 $\pm$ 0.0047	0.0134	[0.0045, 0.0246]	12/894	0.0425	0.0184

Table 1: **Table 1. Corpus-supported distractor rates by arm.** Conditional is the distractor-caused CSD rate over retained distractor slots, averaged across three seeds (mean $\pm$ seed-level standard deviation). Pooled weights seeds by retained slots and carries a Wilson 95% interval. Events is the exact flagged-slot count over retained slots. Undecomp. is the undecomposed corpus-supported rate, the sum of the distractor-caused and stem-caused routes, seed-averaged. Soft is the arm-level mean detector support probability over three seeds, which does not quantise. No arm is marked best, since no superiority claim among treatments is supported at this sample size. Control rows are referents, not methods.

whichever route produced the defect, the grounded pool reaches 3.24% and the ungrounded control 0.68%, a ratio of 4.8 against the 9.1 the subset shows. Roughly seven of every ten flagged slots in the grounded pool are stem-caused rather than distractor-caused (2.24% against 1.00%), which means most of this defect is not something a distractor filter could ever remove. That figure alone reorders the practical priorities, since it points at stem specificity rather than at option filtering, and we state it here rather than developing it because three seeds will not support the follow-up. Three measures agreeing on a direction at three seeds is suggestive and no more, and Section 6 gives two independent reasons to hold the grounded-versus-ungrounded reading loosely.

5.2 Seed structure

Table 2 breaks the arms down per seed and exposes the regime that governs every downstream test.

Seed 0 carries most detected events in every arm, seed 1 produces zero events in nine of ten arms, and seed 2 sits between them. This is a property of the shared generation pool rather than of any arm, which is why seed-level standard deviations run at the same order as the means, and why the t -test’s normality assumption holds only loosely on zero-inflated data. The left panel of Figure 4 shows the spread arm by arm.

5.3 Filters against the attrition null

Table 3 reports every paired contrast against the unfiltered grounded pool.

Read as rates alone, this table says that a

grounding-based faithfulness filter roughly doubles the corpus-supported distractor rate, from 1.01% to 2.01%. That is the headline the question anticipated, and it is wrong, or at least it is not what the numbers support. The retention-matched random-drop null raises the rate to 1.79% by discarding items on a coin flip, and its own bootstrap interval also excludes zero. Every filter contrast sits inside the null’s territory. The largest filter-minus-null difference is +0.0022, which is a fifth of the interval width on either arm.

The decisive evidence is arithmetic rather than inferential, and Figure 3 carries it. Filters retain half the pool, so their denominators fall from 894 distractor slots to 447. Their numerators do not move. The unfiltered pool flags 9 slots, and per-option entailment, not-entail polarity and contradiction polarity each flag 9 as well on half the slots. The run recorded counts rather than slot identities, so we claim equality of counts and not that the same nine objects survived, though the arithmetic below does not depend on which is true. The random-drop null flags 8, having removed one flagged slot by chance while removing half of everything. A conditional rate that doubles as the denominator halves is the signature of a gate that discards items without discarding the defect.

Testing the counts directly settles the comparison the rates only gesture at. On a two-proportion test over distractor slots, per-option entailment filtering does not separate from the unfiltered pool (9/894 against 9/447, $z = 1.51$, $p = 0.13$, Fisher exact $p = 0.14$), and it does not separate from the attrition null either (9/447 against 8/447, $z = -0.25$, $p = 0.81$,

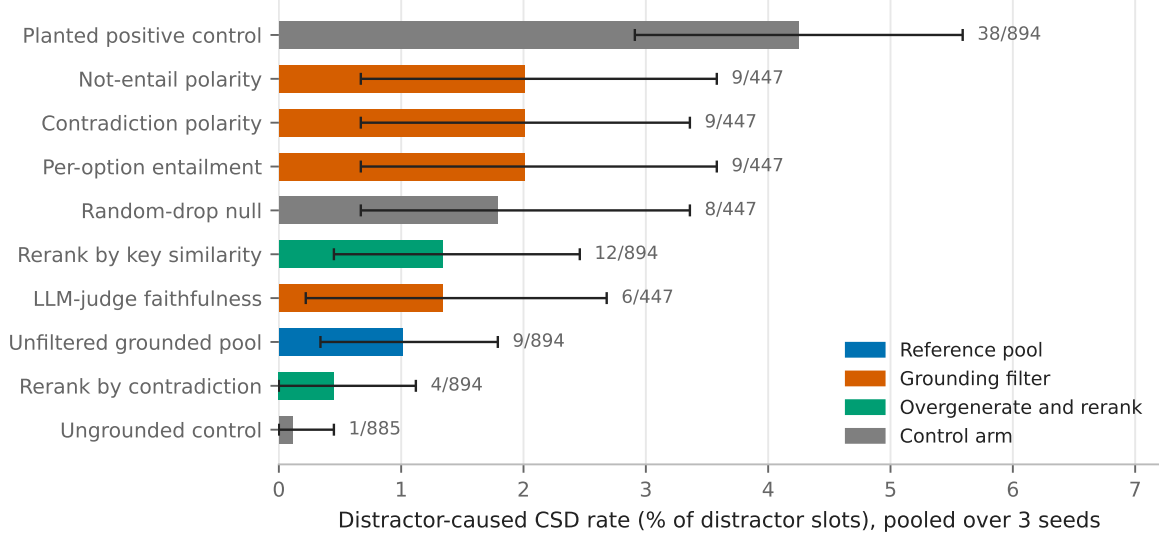


Figure 2: Distractor-caused CSD rate by arm, pooled over three seeds, with Wilson 95% intervals and the exact flagged-slot count beside each bar. The planted positive control separates above every natural arm and the ungrounded control sits below them, while six filter arms and the attrition null compress into a band whose internal ordering the design cannot resolve.

Arm	Seed 0	Seed 1	Seed 2	Mean
Unfilt	0.0200	0.0000	0.0100	0.0100
Ungrnd	0.0034	0.0000	0.0000	0.0011
PerOptEnt	0.0400	0.0000	0.0200	0.0200
NotEntPol	0.0400	0.0000	0.0200	0.0200
ContraPol	0.0400	0.0000	0.0200	0.0200
LLMJudge	0.0267	0.0000	0.0133	0.0133
RRContra	0.0133	0.0000	0.0000	0.0044
RRKeySim	0.0200	0.0102	0.0100	0.0134
NullDrop	0.0333	0.0000	0.0200	0.0178
Planted	0.0533	0.0272	0.0467	0.0424

Table 2: **Table 2. Per-seed conditional CSD rates.** Seed 1 yields zero detected events in nine of ten arms. The three entailment-family filters return identical per-seed vectors, so any test comparing them is degenerate.

Fisher exact $p = 1.00$). A Fisher p of 1.00 against the null is as close to no evidence as this design can produce. The null itself does not separate from the unfiltered pool (Fisher exact $p = 0.30$). Whatever these filters do to the retained pool, they do not do it to the defect.

One filter is the exception and it deserves separating from the family. The LLM-judge faithfulness filter dropped the flagged-slot count from 9 to 6, removing three where the attrition null removed one, so it is the only one of the four that touched the numerator at all. Its conditional rate still rose, because 6 of 447 exceeds 9 of 894, which is exactly the trap this section is about. On the counts it separates from neither the unfiltered pool (Fisher exact $p = 0.59$) nor per-option entailment filtering ($p = 0.60$), so three removed events out of nine is a direction and not a result. It is worth saying which

of the four filters that direction belongs to, because a larger run should test the generative verifier rather than the NLI family, and the rate table on its own points the other way.

The one intervention that moves the numerator is overgenerate-and-rerank by contradiction, which flags 4 slots out of the full 894 rather than 9, with a bootstrap interval on the difference of $[-0.0112, -0.0000]$ that touches zero at its upper edge. It also carries by far the lowest mean distractor support probability of any arm at 0.0094, roughly a quarter of the reference pool’s. Selecting distractors that the evidence contradicts is a different operation from removing items whose distractors the evidence supports, and only the first has anywhere to move the defect to. At three seeds we report this as the one direction worth pursuing, not as an established reduction.

Comparison	Mean diff	<i>t</i>	<i>p</i>	Cohen’s <i>d</i>	Bootstrap 95% CI
Planted vs Unfilt	+0.0324	11.703	0.0072	6.756	[0.0213, 0.0447]
NullDrop vs Unfilt	+0.0078	1.941	0.1917	1.121	[0.0014, 0.0162]
PerOptEnt vs Unfilt	+0.0100	1.732	0.2254	1.000	[0.0037, 0.0177]
NotEntPol vs Unfilt	+0.0100	1.732	0.2254	1.000	[0.0040, 0.0184]
ContraPol vs Unfilt	+0.0100	1.732	0.2254	1.000	[0.0037, 0.0174]
LLMJudge vs NLI-flag ref	+0.0033	1.732	0.2254	1.000	[-0.0034, 0.0106]
RRKeySim vs Unfilt	+0.0034	1.000	0.4226	0.577	[-0.0045, 0.0112]
Ungrnd vs Unfilt	-0.0089	-1.836	0.2077	-1.060	[-0.0168, -0.0022]
RRContra vs Unfilt	-0.0056	-1.890	0.1994	-1.091	[-0.0112, -0.0000]

Table 3: **Table 3, paired two-sided *t*-tests against the unfiltered grounded pool over the shared seed index ($n = 3$, $df = 2$).** A positive mean difference means a higher CSD rate than the reference. CI is the item-level bootstrap interval on the same difference. The LLM-judge arm is compared against an independent NLI-flag reference, as its own filter consumes the LLM-judge signal.

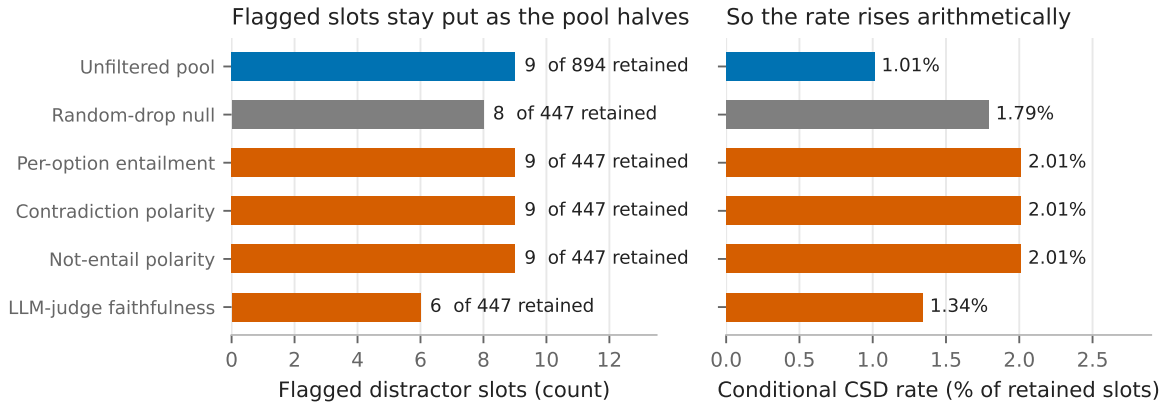


Figure 3: Left panel, flagged distractor slots as a count for the unfiltered pool, the attrition null and the four grounding filters, with the retained denominator printed beside each bar. Right panel, the conditional rate those two numbers produce. The entailment-family filters remove none of the nine flagged slots while discarding half the pool, so their rate doubles arithmetically.

5.4 Detector diagnostics

Judge precision inside the highest lexical-overlap decile is 0.9630 over 160 calibration instances, so the detector’s confident firings in the region where overlap is most likely to mislead are almost all upheld. Overlap explains 21.71% of the variance in the entailment score ($R^2 = 0.2171$), which places most of the signal outside the confound, and the judge’s calibration AUROC is 0.884 over 1,600 pairs with 401 positives. All of that is machine-estimated. No human adjudicated any flagged option, and precision inside a decile says nothing about recall on the rare class it is trying to catch (Powers, 2020).

The right panel of Figure 4 carries a diagnostic that qualifies the control comparison rather than supporting it. Averaged over seeds, the corpus fails to support 61.1% of the grounded arm’s keys and 95.6% of the ungrounded arm’s. The ungrounded generator is writing items about material the re-

trieved passage largely does not contain, so a judge scoring its options against that passage has little to find in any slot, key or distractor. Part of the ungrounded arm’s low CSD rate is therefore a floor rather than a finding.

5.5 Replication and non-replication

A replication on 200 OpenBookQA items per arm gives 0.0033 for the grounded pool and 0.0034 for the ungrounded control. The grounding gap does not reproduce. We report this alongside the main result rather than in a footnote, because it is the strongest single piece of evidence against the reading that retrieval causes the defect, and because a 200-item replication on one corpus is itself far too small to settle the question in either direction.

Two further negative results belong here. The declared per-option threshold grid, $\tau \in \{0.3, 0.5, 0.7, 0.9\}$, admitted no operating point, since every threshold left fewer than the 30 items the enrichment analysis required, so no per-

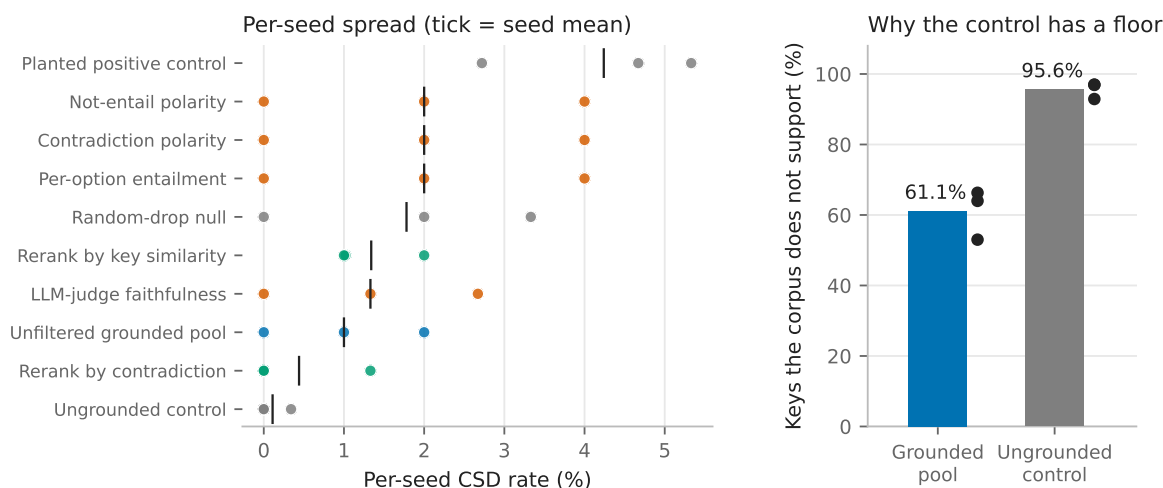


Figure 4: Left panel, per-seed CSD rate for every arm with the seed mean marked, showing that seed 1 contributes zero events in nine of ten arms. Right panel, the share of keys the corpus does not support in the grounded and ungrounded arms, with per-seed values overlaid.

threshold sensitivity curve exists and none is reported. And the pipeline’s grounding attribution posterior returned 0.0, meaning the run produced no positive evidence attributing flagged slots to the grounding step specifically.

6 Discussion

The clearest signal here is the one the design was built to isolate. The planted control separates from the reference pool at $p = 0.007$ while every filter contrast returns $p > 0.19$, so the instrument has demonstrated sensitivity and the filter comparison is limited by how rare the phenomenon is rather than by a blunt detector. That ordering matters, because a null result from an uncalibrated instrument means nothing at all.

The denominator decomposition is the finding we did not anticipate, and it has the most immediate practical consequence. Across the entailment family and the attrition null alike, conditional rates rise in proportion to the halving of the retained pool, which is what one observes when a gate discards items drawn without regard to the defect. A naive comparison would have read that as evidence that grounding filters *increase* the defect rate, which is the exact claim the research question invited us to make. The parsimonious reading is that these filters neither increase nor decrease the underlying event count, and merely change what the count is divided by. The count-level test puts a number on how little separates a filter from random attrition, since per-option entailment filtering

and the random-drop null return a Fisher exact p of 1.00 against each other on identical denominators. The retention-matched null is what makes the two readings distinguishable, and its absence from published filtering studies means a reported quality gain cannot be attributed to a gate’s criterion rather than to its attrition. Reviews locate the item-generation bottleneck in validation effort (Tan et al., 2025; Falcão et al., 2022). Part of it sits in validation design.

The enumeration account that motivated the corpus choice is a hypothesis this run does not test. Enumeration density was computed over the corpus and recorded at a median of 0.167, and it was never used as an endpoint, so nothing here shows that enumerated passages produce more corpus-supported options than continuous prose does. We keep the account because it explains why a narrow instructional index is the setting to worry about, and we mark it as untested because a reader would otherwise take Section 4.1 for a result.

The stem-caused share is the practical finding we did not go looking for. About seven of every ten flagged slots in the grounded pool survive only under the original stem, which puts them beyond the reach of any distractor-side intervention, because the option is a correct answer to the question as asked. If that share holds on real course material, the fix belongs in stem specificity rather than in option filtering, and a pipeline spending its verification budget on options is spending it on the smaller share of the problem. At three seeds this is a direction and not a recommendation, and Section

7 records why no point estimate is offered for the split.

Two independent reasons argue against reading the grounded-versus-ungrounded gap as evidence that retrieval causes corpus-supported distractors. The replication on a second corpus finds no gap at all. And the ungrounded arm hallucinates 95.6% of its keys relative to the corpus, against 61.1% for the grounded arm, so the arm that shows a low CSD rate is also the arm whose items are largely off-corpus, where a judge scoring against the corpus has nothing to support. The control is structurally clean, since the passage is provably absent from its generation prompt and the evidence set is held fixed for scoring, and it is still confounded, because withholding the passage changes what the items are about as well as how they were written. A cleaner counterfactual would hold item content fixed and vary only the evidence available at scoring time. We did not run that arm.

The circularity finding generalises past quiz generation and is the most transferable observation here. NLI verifiers are the default instrument for auditing retrieval-augmented systems (Laban et al., 2022; Honovich et al., 2022; Es et al., 2024), and the same machinery is increasingly deployed *inside* pipelines as a filter (Lei et al., 2023; Wu et al., 2024). When filter and metric draw on one signal family, an arm’s apparent score partly reflects its own selection rather than the quality of what it kept, and unlike a power deficit this does not dissolve with more items. Work showing that correctness and faithfulness dissociate in attribution (Wallat et al., 2025; Stolfo, 2024) identifies the conceptual half of the problem. The arm design here surfaces the measurement half. The remedy is architectural, since the endpoint has to be computed by a signal the filter cannot see, for instance a QA-based answerability check (Wang et al., 2020; Fabbri et al., 2022) that never touches the entailment model the filters consume. Falsifying the circularity claim would mean showing that filter-versus-null ordering survives under such an independent endpoint.

The statistical half does dissolve with scale, and the operating point is computable. A two-proportion power calculation for a doubling from a 1.0% base rate to a 2.0% alternative, two-sided at $\alpha = 0.05$ with 80% power and slots treated as independent, returns 2,318 distractor slots per arm, which is 773 items at three slots each. That is roughly two and a half times the pool used here on items, and more once clustering within items

and seeds is accounted for, since the independence assumption the calculation rests on is the same one Section 4.3 flags as anti-conservative. Three implications follow for anyone shipping a grounded item generator. A faithfulness score is not an item-validity score, since a maximally faithful distractor can be maximally invalid. Any quality gate reporting a reduction should be benchmarked against a retention-matched null before its criterion is credited, because attrition alone moves a conditional rate. And a dashboard surfacing groundedness as a proxy for assessment quality is reporting a quantity that is logically independent of the one instructors care about.

7 Limitations

- **Power.** At a base rate near 1% on 100 items per seed and 300 per arm, every inter-arm gap spans between one and five flagged slots. No filter contrast reaches significance and no superiority claim among the six treatment arms is made or should be inferred from Table 1. Forty-five pairwise contrasts are available and none would survive multiplicity correction. The seed-level standard deviations reported with the means are dispersion measures, not confidence intervals, and the data are zero-inflated at seed 1, so the *t*-test’s normality assumption holds only approximately.
- **Corpus and the headline generalisation.** The evaluation runs on SciQ support passages and OpenBookQA core facts, not lecture notes or textbook chapters. These supports are shorter and less often enumerative than genuine course material, which plausibly depresses the base rate and therefore the study’s power. The paper’s title question names a course corpus, and we did not run one.
- **The grounding claim does not replicate.** The OpenBookQA replication gives 0.0033 against 0.0034 for the grounded and ungrounded arms, so the grounded-versus-ungrounded gap appears on one corpus and not on the other. No causal claim about retrieval is supported by this run.
- **Control confound.** The ungrounded arm’s 95.6% key hallucination rate means its low CSD rate is partly a floor effect, as Section 6 sets out. The arm is a valid manipulation

of generation context and a poor isolation of retrieval’s causal role.

- **Detector calibration.** The planted control shows the detector separates constructed positives from the reference pool, but injection rate, recovered planted fraction and the resulting recall ratio were not reported as endpoints, so no recall-corrected prevalence exists and every natural-prevalence figure is observed rather than corrected. Judge precision is machine-estimated with no human adjudication. The stem-caused share reported in Section 5.1 and Section 6 is a seed-averaged ratio over between one and five flagged slots per seed, so it carries no usable interval, which is why no stem-rewriting recommendation is advanced from it. The enumeration-density moderator was recorded and never tested as an endpoint.
- **Configuration and unreported sensitivity.** One generator, one entailment model, one retriever, one corpus and a fixed top-50% retention rule mean every mechanism claim is confounded with a single configuration. The threshold grid was declared but admitted no operating point, so per-threshold sensitivity is unavailable. Four-bit quantisation failed at run time, the NLI judge ran on CPU, and 2,460 of 15,951 scored pairs were truncated at the 512-token limit, which can only lose evidence. Six items in the contradiction-rerank arm at seed 0 had fewer than the required candidates after repair and were padded with their original distractors. Parse failure rates reached 2% in one arm-seed.
- **Validation and provenance.** Corpus contamination of the generator’s pretraining is not excluded. Earlier repair rounds of the same pipeline completed fewer arm-seeds, whereas the run summarised here completed all thirty, and replicate variance from decoding non-determinism and retrieval tie-breaking is the largest uncontrolled term.

8 Conclusion

We named corpus-supported distractors, options that sit in a distractor slot while the retrieved evidence establishes them as correct answers to their own stem, showed that passage-only faithfulness metrics are logically incapable of detecting them,

and built a stem-conditioned detector evaluated inside a ten-arm control design. The detector separates a planted positive control from the reference pool at $p = 0.007$, which is the sensitivity floor everything else stands on. On a science-exam corpus standing in for course material, the corpus supports a wrong option on 3.24% of grounded distractor slots against 0.68% of ungrounded ones, and 1.00% against 0.11% once the stem-underdetermination route is subtracted, a gap three seeds cannot certify and a second corpus does not reproduce. Four grounding-based faithfulness filters are indistinguishable from a retention-matched attrition null on rates and on counts alike, and counting rather than dividing shows the three entailment-family filters left the flagged-slot count at nine while discarding half the pool.

Three directions follow from what this run could and could not settle. An endpoint computed from a signal family the filters cannot see, such as a QA-based answerability check, would break the filter-metric circularity at any sample size. Pools of roughly 773 items per arm, or 2,318 distractor slots, analysed with item-level pairing on a shared generation pool, would settle the filter question statistically. And an evaluation on genuine course material, where enumerated passages are common and where the title question actually lives, would test both the enumeration account and the stem-caused share in the setting they were hypothesised for.

9 Statements

9.1 Ethics

No human subjects were involved and no personal data was processed. The corpus consists of public science-exam passages. The work studies a defect in machine-generated assessment items, and the practical recommendation it supports is that grounding scores should not be used as evidence of item validity in graded settings.

9.2 AI usage disclosure

AI was used to rephrase some statements. All technical content and details were created and written by the author.

9.3 Author contributions

Mohamed Farag contributed conceptualisation, methodology, formal analysis, investigation, software audit, and writing of both the original draft

and the revision.

9.4 Conflict of interest

The author declares no competing interests. The research question arises from an education product under development at AGAI, and no result here is favourable to that product.

9.5 Funding

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References

- Elaf Alhazmi, Quan Z. Sheng, Wei Emma Zhang, Munazza Zaib, and Ahoud Alhazmi. 2024. [Distractor generation in Multiple-Choice tasks: A survey of methods, datasets, and evaluation](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- Halim Wildan Awalurahman and Indra Budi. 2024. [Automatic distractor generation in multiple-choice questions: a systematic literature review](#). *PeerJ Computer Science*.
- Semere Kiros Bitew, Johannes Deleu, Chris Develder, and Thomas Demeester. 2025. [Distractor generation for Multiple-Choice questions with Predictive prompting and Large language models](#). *Communications in Computer and Information Science*.
- Semere Kiros Bitew, Amir Hadifar, Lucas Sterckx, Johannes Deleu, Chris Develder, and Thomas Demeester. 2024. [Learning to reuse distractors to support Multiple-Choice question generation in education](#). *IEEE Transactions on Learning Technologies*.
- Alexander Braylan, Omar Alonso, and Matthew Lease. 2022. [Measuring annotator agreement generally across complex structured, multi-object, and free-text annotation tasks](#). In *Proceedings of the ACM Web Conference 2022*.
- Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. 2023. [ChatEval: towards better LLM-based evaluators through Multi-Agent debate](#). *Preprint*, arXiv:2308.07201. Preprint, not peer reviewed.
- Jiawei Chen, Hongyu Lin, Xianpei Han, and Le Sun. 2024. [Benchmarking large language models in Retrieval-Augmented generation](#). *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Neil Chiavaroli. 2017. [Negatively-Worded multiple choice questions: An avoidable threat to validity](#). *Practical Assessment, Research & Evaluation*.
- Ho-Lam Chung, Ying-Hong Chan, and Yao-Chung Fan. 2020. [A BERT-based distractor generation scheme with multi-tasking and negative answer training strategies](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*.
- Jacob Doughty, Zipiao Wan, Anishka Bompelli, Jubahed Qayum, Taozhi Wang, Juran Zhang, Yujia Zheng, Aidan Doyle, Pragnya Sridhar, Arav Agarwal, Christopher Bogart, Eric Keylor, Can Kultur, Jaromir Savelka, and Majd Sakr. 2024. [A comparative study of AI-Generated \(GPT-4\) and human-crafted MCQs in programming education](#). In *Proceedings of the 26th Australasian Computing Education Conference*.
- Esin Durmus, He He, and Mona Diab. 2020. [FEQA: a question answering evaluation framework for faithfulness assessment in abstractive summarization](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.
- Shahul Es, Jithin James, Luis Espinosa Anke, and Steven Schockaert. 2024. [RAGAs: automated evaluation of retrieval augmented generation](#). In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*.
- Alexander Fabbri, Chien-Sheng Wu, Wenhao Liu, and Caiming Xiong. 2022. [QAFactEval: improved QA-Based factual consistency evaluation for summarization](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*.
- Filipe Falcão, Patrício Costa, and José M. Pêgo. 2022. [Feasibility assurance: a review of automatic item generation in medical assessment](#). *Advances in Health Sciences Education*.
- Wenqi Fan, Yajuan Ding, Liangbo Ning, Shijie Wang, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. 2024. [A survey on RAG meeting LLMs: towards Retrieval-Augmented large language models](#). In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.
- Fatema Tuj Johora Faria, Mukaffi Bin Moin, M. F. Mridha, and Jubayer Al Mahmud. 2026. [TeachMateGPT: a Multi-Agent Knowledge-Grounded framework for pedagogical assessment generation from science curriculum materials](#). *Preprint*, arXiv:2608.13708. Preprint, not peer reviewed.
- Wanyong Feng, Jaewook Lee, Hunter McNichols, Alexander Scarlatos, Digory Smith, Simon Woodhead, Nancy Ornelas, and Andrew Lan. 2024. [Exploring automated distractor generation for math multiple-choice questions via large language models](#). In *Findings of the Association for Computational Linguistics: NAACL 2024*.
- Nigel Fernandez, Alexander Scarlatos, Wanyong Feng, Simon Woodhead, and Andrew Lan. 2024. [DiVERT: distractor generation with variational errors represented as text for math multiple-choice questions](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and

- Haofen Wang. 2023. [Retrieval-Augmented generation for large language models: A survey](#). *Preprint*, arXiv:2312.10997. Preprint, not peer reviewed.
- Sreekanth Gopi, Devananda Sreekanth, and Nasrin Dehbozorgi. 2024. [Enhancing engineering education through LLM-Driven adaptive quiz generation: A RAG-Based approach](#). In *2024 IEEE Frontiers in Education Conference (FIE)*.
- Christian Grévisse, Maria Angeliki S. Pavlou, and Jochen G. Schneider. 2024. [Docimological quality analysis of LLM-Generated multiple choice questions in computer science and medicine](#). *SN Computer Science*.
- Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kukliansy, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. 2022. [TRUE: re-evaluating factual consistency evaluation](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*.
- Or Honovich, Leshem Choshen, Roei Aharoni, Ella Neeman, Idan Szpektor, and Omri Abend. 2021. [Q2: Evaluating factual consistency in Knowledge-Grounded dialogues via question generation and question answering](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*.
- Hui Huang, Xingyuan Bu, Hongli Zhou, Yingqi Qu, Jing Liu, Muyun Yang, Bing Xu, and Tiejun Zhao. 2025. [An empirical study of LLM-as-a-Judge for LLM evaluation: Fine-tuned judge model is not a general substitute for GPT-4](#). In *Findings of the Association for Computational Linguistics: ACL 2025*.
- Wojciech Kryscinski, Bryan McCann, Caiming Xiong, and Richard Socher. 2020. [Evaluating the factual consistency of abstractive text summarization](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Archana Praveen Kumar, Ashalatha Nayak, Manjula Shenoy K, Chaitanya, and Kaustav Ghosh. 2024. [A novel framework for the generation of multiple choice question stems using semantic and Machine-Learning techniques](#). *International Journal of Artificial Intelligence in Education*.
- Yavuz Selim Kiyak, Abdullah Bedir Kaya, and Emre Emekli. 2026. [Validity of AI-generated multiple-choice questions in medical education: a systematic review](#). *Postgraduate Medical Journal*.
- Philippe Laban, Tobias Schnabel, Paul N. Bennett, and Marti A. Hearst. 2022. [<scp>SummaC</scp>: Re-Visiting NLI-based models for inconsistency detection in summarization](#). *Transactions of the Association for Computational Linguistics*.
- Matthias Carl Laupichler, Johanna Flora Rother, Ilona C Grunwald Kadow, Seifollah Ahmadi, and Tobias Raupach. 2023. [Large language models in medical education: Comparing ChatGPT- to Human-Generated exam questions](#). *Academic Medicine*.
- Yukyung Lee, Joonghoon Kim, Jaehee Kim, Hyowon Cho, Jaewook Kang, Pilsung Kang, and Najoung Kim. 2025. [CheckEval: a reliable LLM-as-a-Judge framework for evaluating text generation using check-lists](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*.
- Deren Lei, Yaxi Li, Mengya Hu, Mingyu Wang, Vincent Yun, Emily Ching, and Eslam Kamal. 2023. [Chain of natural language inference for reducing large language model ungrounded hallucinations](#). *Preprint*, arXiv:2310.03951. Preprint, not peer reviewed.
- Haitao Li, Qian Dong, Junjie Chen, Huixue Su, Yujia Zhou, Qingyao Ai, Ziyi Ye, and Yiqun Liu. 2024. [LLMs-as-Judges: a comprehensive survey on LLM-based evaluation methods](#). *Preprint*, arXiv:2412.05579. Preprint, not peer reviewed.
- Qian Liu, Navé Wald, Chandima Daskon, and Tony Harland. 2023a. [Multiple-choice questions \(MCQs\) for higher-order cognition: Perspectives of university teachers](#). *Innovations in Education and Teaching International*.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023b. [G-Eval: NLG evaluation using gpt-4 with better human alignment](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.
- Kaushal Kumar Maurya and Maunendra Sankar Desarkar. 2020. [Learning to distract: A hierarchical Multi-Decoder network for automated generation of long distractors for Multiple-Choice questions for reading comprehension](#). In *Proceedings of the 29th ACM International Conference on Information & Knowledge Management*.
- Dennis Menze, Slaviša Radović, and Niels Seidel. 2026. [AI-assisted MCQ creation increases item-writing flaws through automation bias: evaluation of a workflow of multiple AI agents with teachers](#). *Frontiers in Computer Science*.
- Pradeesh N, Remya T, MG Thushara, K Arun Krishna, and Pranav V. 2025. [Retrieval-Augmented generation for Multiple-Choice questions and answers generation](#). *Procedia Computer Science*.
- Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. 2024. [RAGTruth: a hallucination corpus for developing trustworthy Retrieval-Augmented language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.
- Jeroen Offerijns, Suzan Verberne, and Tessa Verhoef. 2020. [Better distractions: Transformer-based distractor generation and multiple choice question filtering](#). *Preprint*, arXiv:2010.09598. Preprint, not peer reviewed.

- David M. W. Powers. 2020. [Evaluation: from precision, recall and f-measure to ROC, informedness, markedness and correlation](#). *Preprint*, arXiv:2010.16061. Preprint, not peer reviewed.
- Zhaopeng Qiu, Xian Wu, and Wei Fan. 2020. [Automatic distractor generation for multiple choice questions in standard tests](#). In *Proceedings of the 28th International Conference on Computational Linguistics*.
- Vyas Raina, Adian Liusie, and Mark Gales. 2024. [Is LLM-as-a-Judge robust? investigating universal adversarial attacks on zero-shot LLM assessment](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- Siyu Ren and Kenny Q. Zhu. 2021. [Knowledge-Driven distractor generation for Cloze-Style multiple choice questions](#). *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Assad Ali Rezigalla, Ali Mohammed Elhassan Seid Ahmed Eleragi, Amar Babikir Elhussein, Jaber Alfaifi, Mushabab A. ALGhamdi, Ahmed Y. Al Ameer, Amar Ibrahim Omer Yahia, Osama A. Mohammed, and Masoud Ishag Elkhailifa Adam. 2024. [Item analysis: the impact of distractor efficiency on the difficulty index and discrimination power of multiple-choice items](#). *BMC Medical Education*.
- Russell Richie, Sachin Grover, and Fuchiang (Rich) Tsui. 2022. [Inter-annotator agreement is not the ceiling of machine learning performance: Evidence from a comprehensive set of simulations](#). In *Proceedings of the 21st Workshop on Biomedical Language Processing*.
- Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. 2024. [ARES: an automated evaluation framework for Retrieval-Augmented generation systems](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*.
- Kurt Shuster, Spencer Poff, Moya Chen, Douwe Kiela, and Jason Weston. 2021. [Retrieval augmentation reduces hallucination in conversation](#). In *Findings of the Association for Computational Linguistics: EMNLP 2021*.
- Alessandro Stolfo. 2024. [Groundedness in retrieval-augmented long-form generation: An empirical study](#). In *Findings of the Association for Computational Linguistics: NAACL 2024*.
- Jakub Swacha and Michał Gracel. 2025. [Retrieval-Augmented generation \(RAG\) chatbots for education: A survey of applications](#). *Applied Sciences*.
- Bin Tan, Nour Armoush, Elisabetta Mazzullo, Okan Bulut, and Mark Gierl. 2025. [A review of automatic item generation techniques leveraging large language models](#). *International Journal of Assessment Tools in Education*.
- Liyan Tang, Tanya Goyal, Alex Fabbri, Philippe Laban, Jiacheng Xu, Semih Yavuz, Wojciech Kryscinski, Justin Rousseau, and Greg Durrett. 2023. [Understanding factual errors in summarization: Errors, summarizers, datasets, error detectors](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.
- Marie Tarrant, Aimee Knierim, Sasha K. Hayes, and James Ware. 2006. [The frequency of item writing flaws in multiple-choice questions used in high stakes nursing assessments](#). *Nurse Education in Practice*.
- Aman Singh Thakur, Kartik Choudhary, Venkat Srinik Ramayapally, Sankaran Vaidyanathan, and Dieuwe Hupkes. 2024. [Judging the judges: Evaluating alignment and vulnerabilities in LLMs-as-Judges](#). *Preprint*, arXiv:2406.12624. Preprint, not peer reviewed.
- Andrew Tran, Kenneth Angelikas, Egi Rama, Chiku Okechukwu, David H. Smith, and Stephen MacNeil. 2023. [Generating multiple choice questions for computing courses using large language models](#). In *2023 IEEE Frontiers in Education Conference (FIE)*.
- Prasetya Utama, Joshua Bambrick, Nafise Moosavi, and Iryna Gurevych. 2022. [Falsesum: Generating document-level NLI examples for recognizing factual inconsistency in summarization](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*.
- Jonas Wallat, Maria Heuss, Maarten de Rijke, and Avishek Anand. 2025. [Correctness is not faithfulness in retrieval augmented generation attributions](#). In *Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR)*.
- Alex Wang, Kyunghyun Cho, and Mike Lewis. 2020. [Asking and answering questions to evaluate the factual consistency of summaries](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.
- Di Wu, Jia-Chen Gu, Fan Yin, Nanyun Peng, and Kai-Wei Chang. 2024. [Synchronous faithfulness monitoring for trustworthy Retrieval-Augmented generation](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- Jiayuan Xie, Ningxin Peng, Yi Cai, Tao Wang, and Qingbao Huang. 2022. [Diverse distractor generation for constructing High-Quality multiple choice questions](#). *IEEE/ACM Transactions on Audio, Speech, and Language Processing*.
- Han Cheng Yu, Yu An Shih, Kin Man Law, KaiYu Hsieh, Yu Chen Cheng, Hsin Chih Ho, Zih An Lin, Wen-Chuan Hsu, and Yao-Chung Fan. 2024. [Enhancing distractor generation for Multiple-Choice questions with retrieval augmented pretraining and knowledge graph integration](#). In *Findings of the Association for Computational Linguistics ACL 2024*.

- Hao Yu, Aoran Gan, Kai Zhang, Shiwei Tong, Qi Liu, and Zhaofeng Liu. 2025. [Evaluation of Retrieval-Augmented generation: A survey](#). *Communications in Computer and Information Science*.
- Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. 2023. [AlignScore: evaluating factual consistency with a unified alignment function](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.
- Dongxu Zhang, Varun Gangal, Barrett Lattimer, and Yi Yang. 2024. [Enhancing hallucination detection through Perturbation-Based synthetic data generation in system responses](#). In *Findings of the Association for Computational Linguistics ACL 2024*.
- Xiaorui Zhou, Senlin Luo, and Yunfang Wu. 2020. [Co-Attention hierarchical network: Generating coherent long distractors for reading comprehension](#). *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Leonidas Zotos, Hedderik van Rijn, and Malvina Nissim. 2026. [Corpus prevalence of Multiple-Choice question options](#). *Preprint*, arXiv:2602.17377. Preprint, not peer reviewed.
- Devrim Çavuşoğlu, Seçil Şen, and Ulaş Sert. 2024. [Dis-GeM: distractor generation for multiple choice questions with span masking](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*.